

Nisha V Kumar
Technical Engineering Portfolio

M.S.E. Aerospace Engineering, University of Michigan — December 2026
B.S. Aerospace Engineering, The Ohio State University — 2025

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GE Aerospace Research | Thermal Management Systems

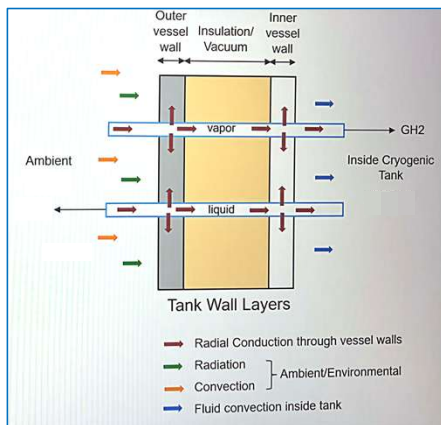
Modeling Transient Two-Phase Cryogenic Hydrogen Tank Behavior for Next Generation Turbofan Fuel Architectures

Objective:

Develop a **transient cryogenic tank model** that delivers liquid H_2 with **engine demand** while recirculating gaseous H_2 for **PI-controlled tank pressurization**, enabling early fuel-system evaluation and design trade studies.

Methods & Tools:

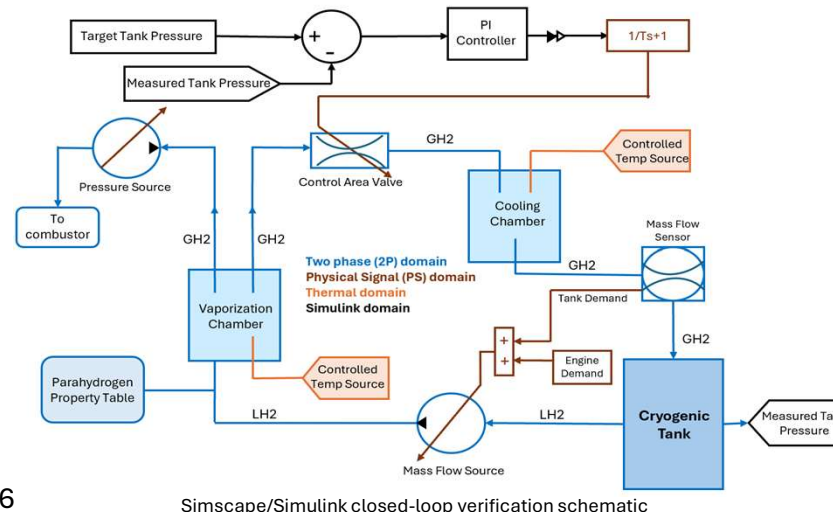
MATLAB | Simscape | Simulink | NIST REFPROP



Conceptual heat-ingress pathways into the 2P cryogenic tank

Technical Approach:

- Developed the **cryogenic tank subsystem**, including three-layer wall heat transfer, material-dependent thermal resistance, ambient heat ingress, and boil-off.
- Built a **standalone system-level verification model** with idealized fuel-delivery, vaporization, cooling, and pressure-control components prior to full architecture integration.
- Implemented **closed-loop H_2 delivery and recirculation**, coupling transient engine demand with liquid withdrawal, vaporization, flow splitting, and PI-controlled gaseous return for tank-pressure regulation.



Outcome:

- Drives early-stage analysis of **two-phase hydrogen behavior** and **transient tank operability**.
- Informs **cryogenic fuel-storage requirements** and **system-level design decisions**.
- Establishes a framework for **fuel-architecture, material, geometry, and mission-level trade studies**.





University of Michigan | Gas Dynamics Imaging Laboratory

Reduced-Order Modeling of Shock-Train Pressure Rise in a Mach 2 Constant-Area Isolator Using a Two-Region Formulation

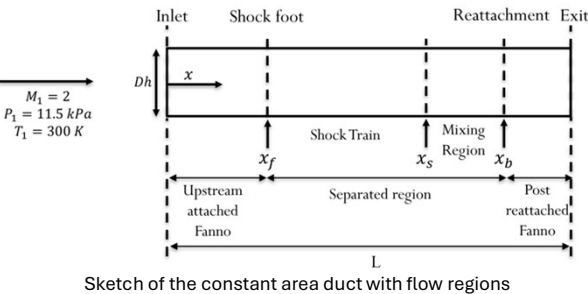
Advisor: Dr. Mirko Gamba | First Author Presentation at 2027 AIAA SciTech Forum

Objective:

Develop a **quasi-one-dimensional reduced-order model** to reproduce shock-train and downstream mixing pressure rise in Mach 2 isolator experiments and establish a **predictive closure for shock-train compression**.

Methods & Tools:

MATLAB | Compressible Flow | Reduced-Order Modeling | Experimental Data



Ortwerth:

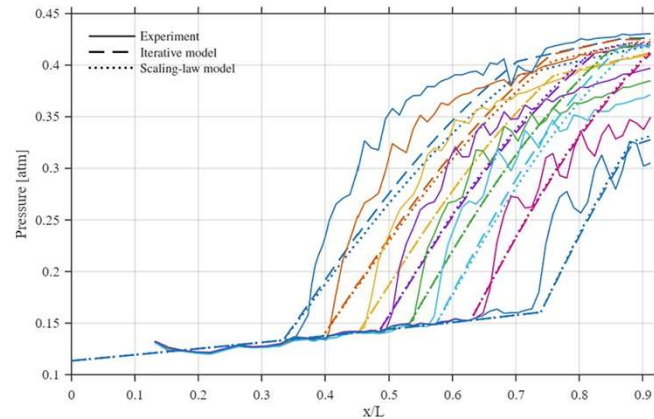
$$\frac{dp}{dx} = \frac{k}{Dh} C f_0 \left(\frac{1}{2} \gamma p M^2 \right)$$

Scaling law:

$$k_1 = 0.070687 \sqrt{\frac{Re_{xf}}{M_f^2 - 1}}$$

Technical Approach:

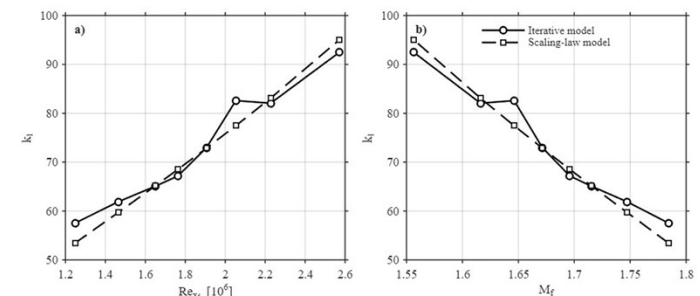
- Coupled **Fanno flow** with a separated-flow formulation in a forward-marching MATLAB model to predict streamwise flow evolution through attached and pseudoshock regions.
- Divided the pseudoshock into distinct **shock-train and mixing regions**, using separate pressure-gradient parameters k_1 and k_2 to capture their differing compression behavior.
- Iteratively calibrated regional parameters against experimental pressure-gradient scaling, then derived a **Mach-Reynolds closure** for k_1 using local shock-foot conditions.



Mach 2 pressure-family comparison for experimental, iterative model, and scaling-law model results

Outcome:

- Reproduced **global pseudoshock pressure-rise behavior** across eight Mach 2 experimental cases.
- Predicted iteratively fitted k_1 values within **2.9% mean difference**, while preserving the modeled pressure distributions.
- Eliminated case-by-case iterative tuning of k_1** using only local shock-foot Mach and Reynolds numbers.



Comparison of iterative and scaling-law shock-region parameter k_1 as functions of shock-foot Reynolds number Re_{xf} and Mach number M_f .





Evaluating AGT1500 Engine Performance Sensitivities & Supporting Compressor Surge Investigation

Objective:

Quantify **AGT1500 low-power engine response** to parameter variation across representative operating conditions to improve cycle-model diagnostics and support **performance certification**, while contributing test-cell analysis to a high-power compressor-surge investigation.

Methods & Tools:

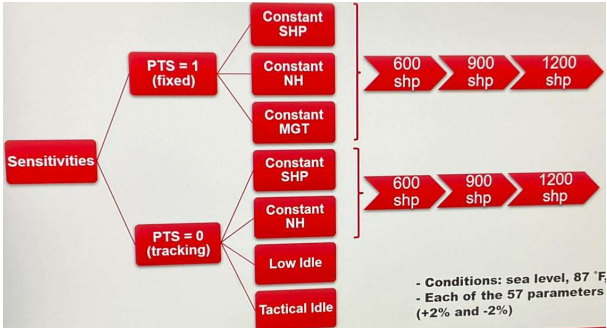
Engine Cycle-Deck Analysis | Sensitivity Analysis | Compressor Performance | Test-Cell Data Reduction

Technical Approach:

- Executed a full-scale **±2% sensitivity study across 57 engine parameters and 17 operating settings**, spanning idle through 1,200 SHP under constant-SHP, NH, and MGT constraints.
- Evaluated **fixed and tracking PTS schedules** to characterize nonlinear engine response and identify discrepancies in low-power cycle-model behavior.
- Supported **1,500-SHP compressor-surge RCCA testing** by recording test-cell data from field-representative power-lever and bleed cases and updating compressor-performance data used for analysis.

Outcome:

- Expanded low-power sensitivity coverage where prior data did not exist, enabling **full-scale test configuration selection and certification analysis**.
- Improved **cycle-model diagnostics** by identifying nonlinear response and legacy-code discrepancies; delivered recommendations through a final technical report.
- Updated compressor performance data supporting **root-cause analysis and surge-prevention efforts at 1,500 SHP**.



Flowchart of parameter variations for sensitivity study



AGT1500 turboshaft engine | M1 Abrams
Publicly available image

Terminology
SHP : shaft horsepower
NH : high-pressure compressor speed
MGT : measured gas temperature
PTS : power turbine stator
RCCA : root cause corrective action



The Ohio State University Rocketry Team | Buckeye Space Launch Initiative

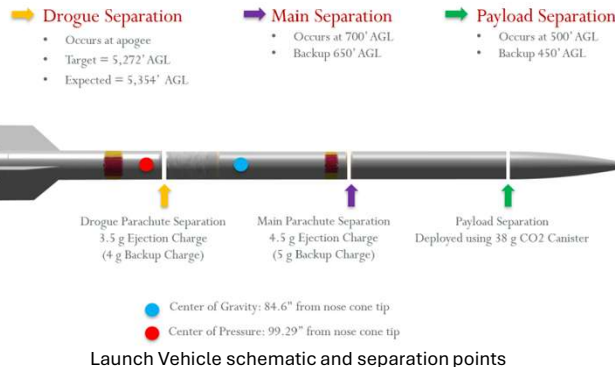
Leading Multidisciplinary Rocket Development, Launch Readiness, & Flight Operations

Objective:

Lead **cross-functional vehicle integration, mission safety, and launch readiness** for a 30+ member team developing an L-class launch vehicle and mission payload for NASA Student Launch, supporting the full **design-build-test-flight lifecycle**.

Methods & Tools:

FMEA | Requirements Verification | Launch Operations | Recovery Systems | OpenRocket | Root-Cause Analysis



Technical Approach:

- Led **system-level risk and requirements verification** across vehicle, payload, recovery, and flight operations using FMEA, personnel/environmental hazard analysis, project-risk tracking, and compliance verification.
- Developed and maintained **launch-readiness procedures and checklists** for hardware packing, subsystem preparation, vehicle integration, recovery/ejection testing, field operations, and post-flight inspection.
- Coordinated **30+ engineers across structures, payload, flight systems, and testing** through design reviews, integration, launch readiness, field execution, and post-flight root-cause analysis.



Successful recovery configuration

Outcome:

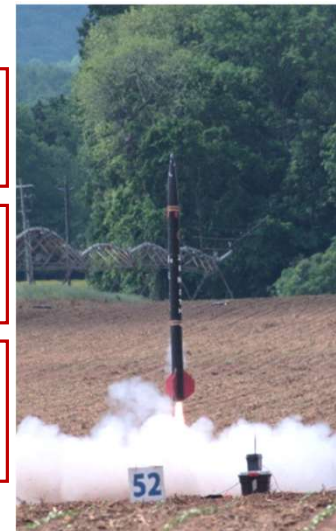
Converted risk analyses and post-flight findings into updated procedures, corrective actions, and launch-readiness controls for subsequent operations.

Program Impact

▲ **24 pts above previous year**
NASA Safety Review Score
31/78 → 50/78

▲ **8.8 pts above previous year**
Major Milestone Review Score
461/750 → 528/750

100% across both years
Flight & Recovery Score
Successful competition launch and recovery



Final competition launch at Huntsville, AL

Deputy Project Manager & Safety Officer | May 2024 – May 2025